



Synergistic anti-liver fibrosis actions of total astragalus saponins and glycyrrhizic acid via TGF- β 1/Smads signaling pathway modulation



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ABSTRACT

Ethnopharmacological relevance: Huangqi decoction (HQD) is a well-known traditional Chinese herbal formulation. It is an effective treatment for consumptive disease and chronic liver diseases. It consists of Radix Astragali (*Astragalus membranaceus* (Fisch.) Bge. Root, Huangqi) and Radix Glycyrrhizae (*Glycyrrhiza uralensis* Fisch., root and rhizome, Gancao). Total astragalus saponins (AST) is a main component of Radix Astragali and glycyrrhizic acid (GA) is a main component of Radix Glycyrrhizae. Our primary results showed that the combination of AST and GA had an obvious synergistic effect in reducing liver collagen deposition and decreasing serum alanine aminotransferase (ALT) activity in dimethylnitrosamine (DMN)-induced liver fibrosis.

Aim of the study: Through *in vivo* and *in vitro* experiments, we aimed at investigating the key anti-fibrosis signal pathway TGF- β 1/Smads to further explore the synergistic mechanism of AST and GA.

Material and methods: Two hepatic fibrosis animal models, bile duct ligation-induced (BDL) and DMN-induced, were utilized. Rats were treated orally with AST, GA or AST/GA, with the effects evaluated via liver histopathology, hydroxyproline (Hyp) levels, and α -SMA expression. In the hepatic stellate cell line JS-1, cells were treated with AST/GA for 24 h, followed by a cell viability assessment using Cell Counting Kit-8 (CCK-8) and Real-time PCR and Western blot analysis of α -SMA, Coll and TGF- β 1/Smads signaling pathway related components.

Results: The AST/GA combination attenuated liver tissue inflammation, collagen deposition, Hyp levels, and α -SMA expression in both BDL-and DMN-stimulated hepatic fibrosis rats. *In vitro* results showed that the AST/GA combination significantly inhibited JS-1 cell viability, significantly suppressed α -SMA, Coll, TGF- β 1, Smad2 and Smad3 mRNA and protein expression, as well reduced p-Smad2/3. Compared with AST or GA treatment alone, the AST/GA combination significantly reduced Smad3 mRNA expression levels and TGF- β 1, Smad3, and p-Smad2/3 protein levels.

Conclusions: AST and GA synergistically alleviated both BDL-and DMN-induced hepatic fibrosis via TGF- β 1/Smads signaling pathway inhibition in hepatic stellate cells.

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Abbreviations: ALT, alanine aminotransferase; α -SMA, α -smooth muscle actin; AST, total astragalus saponins; BDL, bile duct ligation; DMN, dimethylnitrosamine; ECM, extracellular matrix; GA, glycyrrhizic acid; GAPDH, glyceraldehyde-3-phosphate dehydrogenase; HSCs, hepatic stellate cells; Hyp, hydroxyproline; LDH, lactate dehydrogenase; TGF- β 1, transforming growth factor- β 1; TGF- β RI, transforming growth factor- β receptor I; UDCA, ursodeoxycholic acid

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1. Introduction

Liver fibrosis is characterized by excessive deposition and qualitative extracellular matrix (ECM) changes in the liver, with these changes associated with general liver wound-healing regardless of origin (Pellicoro et al., 2014). Advanced hepatic fibrosis can lead to liver architecture distortions and the formation of fibrous septa, which contains pseudolobules or nodules, and then eventually develops into liver cirrhosis, which is a kind of irreversible and incurable liver disease. Therefore, attenuating or reversing hepatic fibrosis is a crucial strategy for preventing liver cirrhosis.

During the development of hepatic fibrosis, quiescent hepatic stellate cells (HSCs) are activated to generate α -smooth muscle actin (α -SMA) positive myofibroblast like cells, which are the main contributor to the increased ECM (Delire et al., 2015). During HSC activation, transforming growth factor- β 1 (TGF- β 1) expression, which is the most potent fibrogenic factor, is significantly increased, thus indicating that it serves as an autocrine or paracrine positive regulator for ECM production (Yoshida et al., 2014).

In HSCs TGF- β 1 signaling, transforming growth factor- β receptor I (TGF- β RI) binds its ligand which leads to receptor-activation of Smads (Smad2 and Smad3) via direct serine phosphorylation. This activation then induces Smad2 and Smad3 association with Smad4 and the formed complex translocates into the nucleus where the transcription of specific genes, such as collagens, are impacted (Tahashi et al., 2002). Therefore, the inhibition of HSC TGF- β 1 signaling is considered an effective strategy for the prevention or treatment of liver fibrosis.

While intense research has led to considerable improvements in the understanding of liver fibrosis pathogenesis, effective anti-fibrotic therapies are still lacking. The traditional Chinese herbal formulation, Huangqi decoction (HQD) was first described in Prescriptions of the Bureau of Taiping People's Welfare Pharmacy in the Song Dynasty (AD1078). It is composed of two Chinese herbs, Radix Astragali (*Astragalus membranaceus* (Fisch.) Bge. Root, Huangqi) and Radix Glycyrrhizae (*Glycyrrhiza uralensis* Fisch., root and rhizome, Gancao). It is used in china to clinically treat patients with syndromes such as insufficient deficiency, fatigue, heart palpitations, mouth thirst and sallow complexion, and It has been routinely used for chronic liver diseases, including liver cirrhosis (Liu et al., 2015). Previously, we showed that HQT could inhibit the hepatic fibrosis progression in dimethylnitrosamine (DMN)-induced hepatic fibrosis models (Liu et al., 2012). In addition, Our studies have indicated that there are 4 effective ingredients (including astragalus saponins, astragalus flavonoids, glycyrrhizae acid, and glycyrrhizae flavonoids) able to inhibit hepatic fibrosis progression in an animal model, with the combination of astragalus saponins and glycyrrhizic acid shown to be more effective than even HQD (Tong et al., 2011). In this study, *in vivo* and *in vitro* experiments were utilized to examine the key anti-fibrotic signal pathway TGF- β 1/Smads to further explore the synergistic mechanism of AST and GA.

2. Materials and methods

2.1. Animals

Wistar male rats (SPF grade), with a weight of 150–180 g, were provided by the Shanghai Experimental Animal Center of the Chinese Academy of Sciences [certificate number: SCXK (Shanghai) 2008–0003] and were raised in the Experimental Animal Center of Shanghai University of Traditional Chinese Medicine. Animals were kept in a constant temperature environment and supplied with chow and water *ad libitum*. This study was approved by the Experimental Animal Ethics Committee of Shanghai University of Traditional Chinese Medicine.

2.2. Cell line

An immortalized mice hepatic stellate cell line (JS-1) and a normal mice hepatocyte cell line (AML-12) were obtained from the cell bank of the Institute of Hepatology of Shanghai University of Traditional Chinese Medicine.

2.3. Drugs

AST (purity: 90%, Batch NO: 60,614) was purchased from Xi'an Honson Biotechnology Co., Ltd. (Xi'an, China), GA (purity: 95%, Batch NO: ZL20141011GCS) was purchased from Nanjing Zelang Pharmaceutical Co., Ltd. (Nanjing, China).

2.4. Reagents

Dimethylnitrosamine(DMN) was purchased from Tokyo Chemical Industry (MAL05#, Tokyo, Japan). A serum ALT test kit was purchased from the Nanjing Institute of Biological Products (20080301#, Nanjing, China). A hydroxyproline (Hyp) standard was purchased from the Nakateyitesuku Company (MIR8282#, Tokyo, Japan). CCK-8 cell viability kit (40203ES80#, Shanghai, China) was purchased from the Yeasen Biotechnology Institute and a LDH cytotoxicity assay kit (C0017#, Haimen, China) was purchased from Beyotime Biotechnology Institute. A total RNA Purification Kit MagExtractor (NPK-201#, OSAKA, Japan), ReverTra Ace qPCR-RT Master Mix with gDNA remover (FSQ-301#, OSAKA, Japan), SYBR Green Real-time PCR Master Mix (QPK-201#, OSAKA, Japan) were all purchased from TOYOBO company. A BCA protein assay kit (23225#, Thermo, California, American) and RIPA lysate (P0013B#, Beyotime Biotechnology Institute, Haimen, China) were also purchased. A protease inhibitor cocktail (B14001#, Houston, American) and phosphatase inhibitor cocktail (B15001#, Houston, American) were bought from the Selleck Company. All antibodies used for Western blotting are listed in Table 1 and Real-time PCR primers are listed in Table 2 and were synthesized by Sangon Biotech.

2.5. Bile duct ligation hepatic fibrosis animal model

Wistar male rats were initially randomized into two groups: a sham control group (n=8) and a model group (n=63). Under 2% pentobarbital sodium anesthesia (2 ml/kg, intraperitoneally), the common bile duct was exposed through a midline abdominal incision, with a stiffening agent injected retrograding into the hilum. The section between the ligatures was then cut, the incision was closed, and the animals were allowed to recover for 4 weeks. After one week of surgery, the model rats were randomly divided into five groups: AST/GA combination group [n=12; AST (164 mg/kg/d) + GA (48 mg/kg/d) PO]; AST group [n=12; AST (164 mg/kg/d) PO]; GA group [n=12; GA (48 mg/kg/d) PO]; and ursodeoxycholic acid group [n=12; UDCA (80 mg/kg/d) PO]. All drugs were dissolved in distilled water prior to use and treatments were administered by gavage at 10 ml/kg body weight via intragastric administration daily for a total of 3 weeks. Rats in the sham and model control groups (n=15) were administrated with the same volume of distilled water.

2.6. DMN induced hepatic fibrosis model

Wister male rats (n=50) were randomly allocated to five

Table 1
Western blot antibodies.

Anti-body	Genus	Company	Article number	Dilution ratio
α -SMA	Rabbit	Abcam	Ab124964	1:1000
Coll	Rabbit	Signal way	29155	1:500
TGF- β 1	Rabbit	Abcam	Ab92486	1:400
Smad2	Rabbit	Abcam	Ab33875	1:1000
Smad3	Rabbit	Abcam	Ab40854	1:2500
p-Smad2/3	Rabbit	Santa Cruz	Sc-11769	1:400
GAPDH	Rabbit	Cell Signaling	2118	1:1000

Table 2
Real-time PCR primer sequences.

Gene		Primer sequence	Note
α -SMA	Forward	AATGGCTCTGGGCTCTGTAA	SYBR
	Reverse	TCTCTTGCTCTGGGCTTCAT	
Coll	Forward	TGACTGGAAGAGCGGAGAGT	SYBR
	Reverse	GACGGCTGAGTAGGGAACAC	
TGF- β 1	Forward	ATTCTGGCGTTACCTTGG	SYBR
	Reverse	AGCCCTGTATCCGTCCT	
Smad2	Forward	TTTGGGCGAAGAGATAGC	SYBR
	Reverse	TGTGTATGTGGCGTGTG	
Smad3	Forward	CGCTCTCTCTGCTTTGTGT	SYBR
	Reverse	ATCCTGGTGTGTGGCTTCT	
GAPDH	Forward	AAGGTCATCCATGACAACTTGGC	SYBR
	Reverse	ACAGTCTTCTGGGTGGCAGTGAT	

groups: normal control group, DMN group, AST/GA combination+DMN group, AST+DMN group and GA+DMN group, with 10 rats per group. Liver fibrosis was induced by intraperitoneal injection with 0.5% DMN in saline (2 ml/kg body weight) for three consecutive days per week for up to four weeks. AST and GA were administered by oral gavage once a day for two weeks after two weeks of DMN treated, with the same doses and administration methods employed in the bile duct ligation-induced (BDL) model. Rats in the normal control group (n=10) and DMN control group were treated with the same volume of distilled water.

2.7. Animal specimen collection

At the end of the experiment, rats were sacrificed by anesthesia, liver and serum samples were collected. Liver samples were sectioned (4 μ m) and stained with Hematoxylin–eosin (H&E) or sirius red to assess hepatic collagen deposition and serum ALT levels were determined by reagent kit. Liver Hyp levels were determined as previously described (Jamall et al., 1981).

2.8. Immunohistochemistry

Tissue α -SMA expression levels were determined using the Strept Avidin-Biotin-Peroxidase Complex (SABC) method according to the SABC kit manual. Briefly, paraffin embedded sections were dewaxed with water, endogenous peroxidase were inactivated using 3% H₂O₂, antigens were heat repaired and samples were blocked with 5% BSA. Next, primary antibody was added and allowed to incubate overnight at 4 °C. Biotinylated secondary antibody was then added followed by SABC-peroxidase, with each incubated for 30 min at 37 °C. Samples were visualized with DAB, mounted, photographed, and analyzed using an optical microscope.

2.9. Cell culture and grouped

JS-1 cells were maintained in DMEM medium supplemented with 10% fetal bovine serum (FBS) and AML-12 cells were maintained in DMEM/F-12 (1:1) medium supplemented with 10% FBS and 40 ng/ml dexamethasone. Both were incubated at 37 °C in a humidified atmosphere containing 5% CO₂. When reaching 80% confluency, the cells were divided into the following five groups: normal control group, TGF- β 1 group, TGF- β 1+AST/GA group, TGF- β 1+AST group, TGF- β 1+GA group and a TGF- β 1+SB-431542 positive control group.

2.10. LDH assay for cytotoxicity

A lactate dehydrogenase (LDH) assay was carried out using a LDH-cytotoxicity assay kit according to the manufacturer. Briefly,

AML-12 were seeded into 96-well plates (8000 cells/well) and treated with different concentrations of AST (2.5–40 μ g/ml) or GA (10–200 μ M) for 24 h. Following treatment, 60 μ L supernatant from each well was transferred to another 96-well plate and 30 μ L LDH assay reagent was added and incubated in the dark at room temperature for 30 min. Five wells without cells were used as blanks and LDH levels were measured via spectrophotometry, with the blank optical density (OD) subtracted from each treatment group OD. Experiments were repeated in triplicate and each treatment was performed in triplicate.

2.11. CCK-8 assay for cell viability

Cell viability was assessed using a cell counting kit-8 (CCK-8) assay, with JS-1 cells seeded in 96-well plates (5000 cells/well) and incubated for 24 h at 37 °C in 5% CO₂. Samples were then incubated with the appropriate treatment (AST/GA, AST, GA, or SB-431542) for another 24 h. Next, 10 μ L CCK-8 reagent was added to each well and incubated for 2 h at 37 °C, with plates read using a microplate reader (BioTek, Hercules, CA, USA) at an absorbance of 450 nm. A mean absorbance value from 3 wells for each treatment was used as the index of cell viability and experiments were repeated in triplicate. Cell viability rate (%)=(absorbance of drugs – absorbance of blank)/(absorbance of control – absorbance of blank) $\times$ 100.

2.12. Real-time PCR

Total RNA was extracted from JS-1 cells (1×10^6) and equal amounts were reverse transcribed into cDNA. The cDNA was amplified using Real-time PCR and the housekeeping gene GAPDH was used as an internal standard to quantify target mRNA levels. The cycling conditions were as follows: Hold: 95 °C for 1 min, followed by 40 cycles at 95 °C for 15 s and 60 °C for 1 min. The experiments were replicated at least three times.

2.13. Western blot

Liver tissue or JS-1 cell lysates were obtained and protein concentrations were determined using a BCA assay. Equal amounts of protein samples were separated by SDS-PAGE and transferred to PVDF membranes which were blocked at room temperature for 1 h in blocking solution. The membranes were then incubated with rabbit monoclonal antibodies against either TGF- β 1, Smad2, Smad3 or p-Smad2/3 overnight at 4 °C, followed by a 1 h incubation at room temperature with HRP-conjugated goat anti-rabbit secondary antibody. The obtained bands were then scanned and analyzed with the Odyssey quantitative western blot near-infrared system and band density was analyzed using Image J software. GAPDH served as an internal control and experiments were performed in triplicate.

2.14. Statistical analysis

One-way ANOVA and LSD test were performed for multiple comparisons. Statistical significance was assumed at $P < 0.05$ and data are expressed as a mean $\pm$ SD.

3. Results

3.1. The effects of AST/GA on liver inflammation and collagen deposition in BDL model rats

Rat deaths during modeling included 8 in the BDL group, 4 in the AST group, 6 in the GA and UDCA group, 3 in the AST/GA

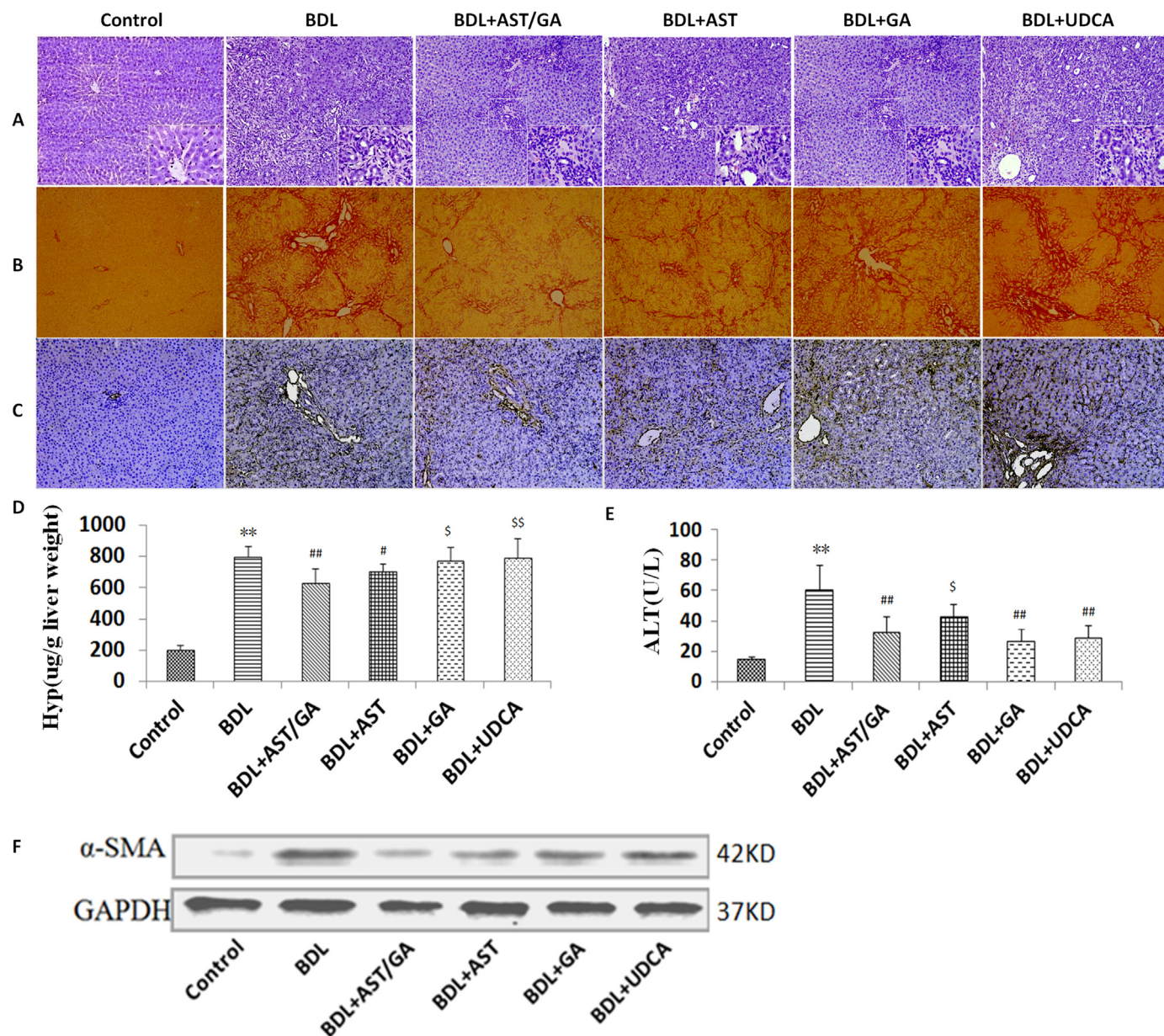


Fig. 1. The effects of AST and GA on liver inflammation and α -SMA expression in BDL rats. (A) H&E staining (200X); (B) sirius red staining (100X); (C) immunohistochemical detection of α -SMA (200X); (D) Hyp levels in liver tissue; (E) serum ALT activity and (F) Western blot detection of α -SMA. * $P < 0.05$ and ** $P < 0.01$ (vs Control); # $P < 0.05$ and ## $P < 0.01$ (vs BDL); \$ $P < 0.05$ and \$\$ $P < 0.01$ (vs BDL+AST/GA).

group. H&E staining showed no apparent architectural abnormalities in the sham group, while in the BDL group, an anadem-like cell shape was observed in the liver tissue, with the portal area in the center and radial extensions reaching the parenchyma. Myofibroblast accumulation and ECM deposition were observed around the anadem shaped cells, with some inflammatory cells infiltrating. The structure of the hepatocytes was normal, while the number of hepatocytes decreased, with an island distribution around the anadem shaped cells noted. The island distribution of the remnant hepatocytes resembled a lobular architecture of different sizes like false lobules and bile duct hyperplasia was noted. Compared with BDL group, the AST/GA and AST groups showed a significant reduction in bile duct hyperplasia and ECM deposition, with no significant change observed in GA or UDCA groups (Fig. 1A).

Sirius red staining showed that there was little collagen fiber deposition in the portal area and central vein walls in the sham group. In the BDL group, extensive ECM deposition was noted

around hyperplastic bile duct epithelial cells, with the portal area being the focal point of the radial extension to liver parenchyma, and little ECM deposition was observed in liver parenchyma. The anadem shape cells and their surrounding collagen fibers formed intervals, joining, encircling and dissecting with adjacent intervals, thus rebuilding virgin hepatic lobules into pseudolobular structures. In the AST/GA and AST groups, bile duct hyperplasia and ECM deposition were significantly reduced compared to the BDL group, with the largest effect noted in the AST/GA relative to the GA and UDCA groups (Fig. 1B).

Immunohistochemical analysis showed that α -SMA was positively expressed in intrahepatic duct and vessel walls, with high expression levels noted in all wall-like structures and hyperplastic bile duct epithelial cells in the model group. While α -SMA expression was in the same locations in the drug intervention groups as in model group, decreased levels were noted, especially in the AST/GA group (Fig. 1C). In the BDL group, α -SMA expression levels were significantly higher than those in the sham group. In

contrast, significantly lower α -SMA expression levels were noted in the AST/GA, AST and GA groups relative to the BDL group, with the lowest levels seen in the AST/GA group (Fig. 1F).

When examining Hyp levels in the liver tissue, levels were significantly increased in the BDL group ($P < 0.01$) relative to the sham group. The AST/GA and AST groups showed a decreased in Hyp levels ($P < 0.05$), with the AST/GA group also having remarkably lower levels than the GA and UDCA groups (Fig. 1D). Additionally, ALT activity was significantly increased in the BDL group relative to the sham group ($P < 0.01$). Alternatively, ALT activity was significantly decreased in the AST/GA, GA and UDCA groups relative to the BDL group ($P < 0.05$; Fig. 1E).

3.2. The effects of AST/GA on liver inflammation and collagen

deposition in DMN model rats

No deaths were noted following modeling and H&E staining showed that the central vein had deviate from the center of hepatic lobule and the portal area architecture was damaged. Furthermore, extensive hepatic centrilobular hemorrhagic necrosis, mononuclear cell infiltration and portal area broadening were noted. Compared with the DMN group, the hepatic parenchymal cells in the AST/GA group showed mild inflammation, while no significant improvement in liver tissue architecture was observed in AST or GA groups (Fig. 2A).

Sirius red staining showed that collagen fiber hyperplasia increased in DMN group, with collagen fibers stretching from the portal area to the hepatic lobule. These collagen fibers divided the hepatic lobule and formed coarse and dense intervals, thus resulting in the loss of normal hepatic lobules and the formation of

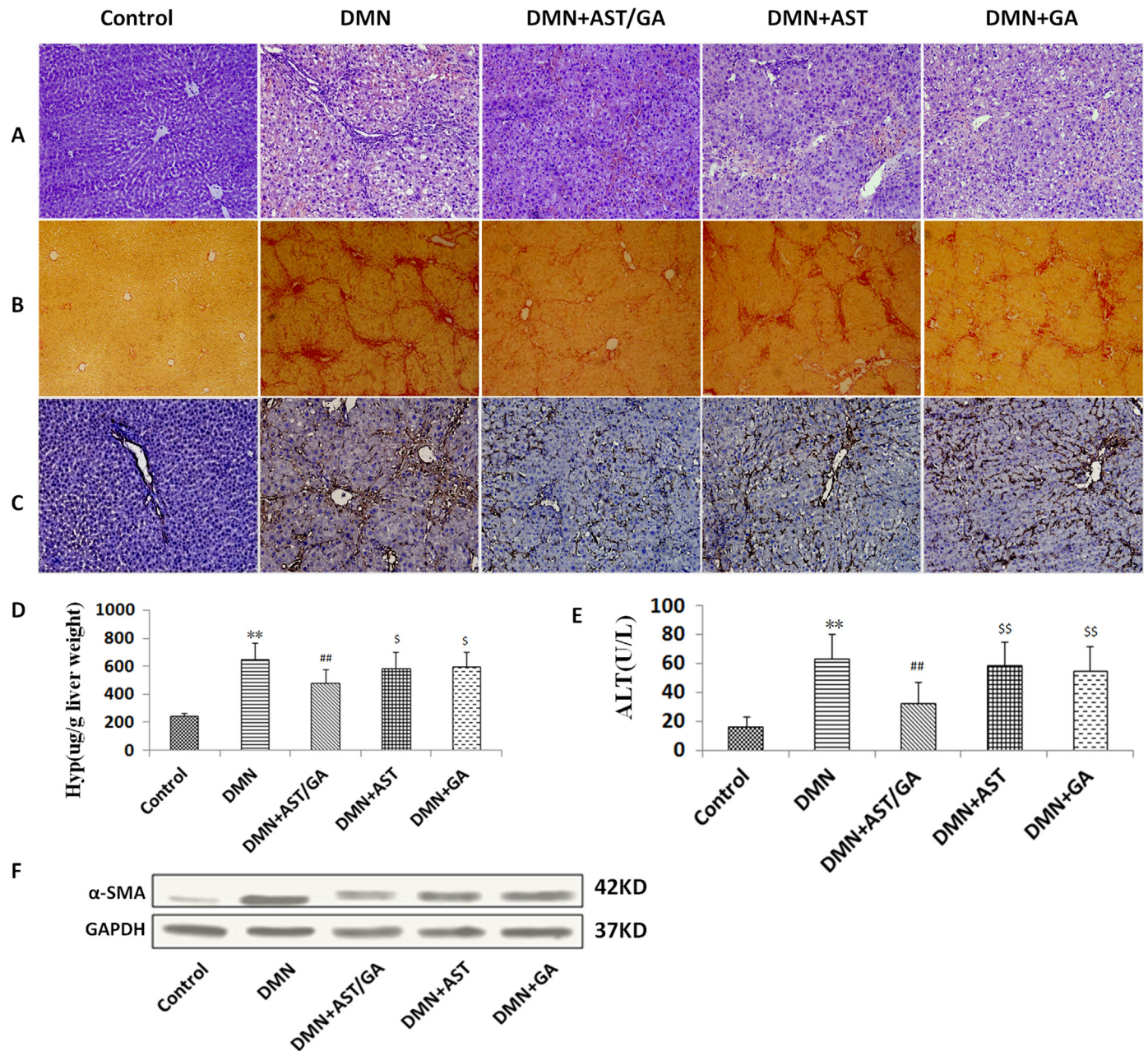


Fig. 2. The effects of AST and GA on liver inflammation and α -SMA expression in DMN rats. (A) H&E staining (200X); (B) sirius red staining (100X); (C) immunohistochemical detection of α -SMA (200X); (D) Hyp levels in the liver tissue; (E) serum ALT activity and (F) Western blot detection of α -SMA. * $P < 0.05$ and ** $P < 0.01$ (vs Control); # $P < 0.05$ and ## $P < 0.01$ (vs DMN); \$ $P < 0.05$ and \$\$ $P < 0.01$ (vs DMN+AST/GA).

some pseudolobuli. The collagen fiber hyperplasia was extenuated by varying degrees following treatment, with the fibrous septum narrow, loose and discrete, and the AST/GA group showed the largest degree of attenuation (Fig. 2B).

When examining liver Hyp levels, the DMN group showed a significant increase ($P < 0.01$; Fig. 2D) relative to the control. Conversely, a significant decreased was noted in the AST/GA group ($P < 0.01$; Fig. 2D) relative to the DMN group. Furthermore, ALT activity was significantly higher in the DMN group relative to the control group ($P < 0.01$; Fig. 2E), while activity levels were significantly decreased in the AST/GA relative to the DMN group ($P < 0.01$; Fig. 2E). The AST/GA group also having remarkably lower levels of Hyp level and ALT activity than the AST and GA groups ($P < 0.01$ or $P < 0.05$).

While α -SMA was only expressed on vessel walls in the control group, it was expressed in the liver parenchyma, except for in the thick and dense fibrous intervals, in the DMN group. Following drug intervention, α -SMA was still expressed in the same positions as in the DMN group, but at reduced levels, especially in the AST/GA group (Fig. 2C). These findings were further substantiated following immunoblotting (Fig. 2F).

3.3. Cytotoxicity of AST and GA on AML-12 cells

LDH, which is an enzyme able to convert lactate to pyruvate, is present in the cytosol, with leakage into the media noted depending on the extent of drug cytotoxicity. Therefore, higher medium LDH levels indicate higher cytotoxicity. When examining various concentrations of AST and GA, no cytotoxic effects were noted (Fig. 3).

3.4. The effects of AST and GA on JS-1 cell viability

A CCK-8 cell viability test showed that AST (40 or 20 μ g/ml) and GA (50 or 100 μ M) groups could significantly reduce JS-1 cell viability when compared with the normal control group, with the highest reduction seen in the AST/GA group (AST: 20 μ g/ml, GA: 50 μ M; Fig. 4). SB-431542 is a specific and potent inhibitor of TGF- β 1 superfamily type I receptors such as, ALK4 (activin receptor-like kinase 4), ALK5 and ALK7, but has no direct effect on components of the ERK, JNK or p38 MAP kinase pathways or associated serum components (Callahan et al., 2002). Following treatment with 10 μ M SB-431542, JS-1 cell viability decreased, thus suggesting that the TGF- β 1 signaling pathway is involved in JS-1 hepatic stellate cell viability.

3.5. Effects of AST and GA on the expression of α -SMA and Collin JS-1 cells

α -SMA was used as a marker for activated HSCs, with levels quantified via Real-time PCR. After treating with various dosages

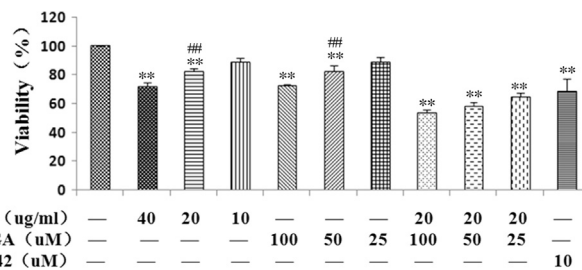


Fig. 4. The effect of AST and GA on JS-1 cell viability. * $P < 0.05$ and ** $P < 0.01$ (vs Control); # $P < 0.05$ and ## $P < 0.01$ [vs AST (20 μ g/ml)+GA (50 μ M)].

of AST (10, 20 or 40 μ g/ml) or GA (25, 50 or 100 μ M), α -SMA mRNA levels were significantly decreased ($P < 0.01$; Fig. 5A), with even lower levels noted in the AST/GA (AST: 20 μ g/ml, GA: 50 μ M) group ($P < 0.01$; Fig. 5A). Additionally, CollmRNA levels were significantly decreased following treatment with AST (20 or 40 μ g/ml) ($P < 0.01$ or $P < 0.05$; Fig. 5B), but no significant differences were noted following treatment with the same GA dosages. The most substantial decrease was noted following treatment with the same AST/GA dosages (Fig. 5B).

Western blot results showed that compared with normal control group, α -SMA and Collexpression was significantly decreased in the AST (20 μ g/ml), AST/GA (AST: 20 μ g/ml, GA: 50 μ M) and positive control (SB-431542) groups ($P < 0.01$; Fig. 5C,D), with the highest reduction noted in the AST/GA group ($P < 0.01$; Fig. 5C,D). Both α -SMA and Collexpression were blocked by the specific TGF- β 1 inhibitor SB-431542. Collectively, these results suggested that the AST/GA combination is the most effective in inhibiting HSC activation by suppressing TGF- β 1.

3.6. Effects of AST/GA on the TGF- β 1/Smads signal pathway in JS-1 cells

TGF- β 1, which can be secreted from HSCs, is the most potent fibrogenic stimulus in HSCs, with elevated TGF- β 1 expression being associated with pathogenesis in various diseases, including liver fibrosis. In HSCs, TGF- β 1 signaling involves Smad2 and Smad3 phosphorylation, thus these were examined via Real-time PCR. TGF- β 1, Smad2 and Smad3 mRNA expression significantly decreased following AST/GA (AST: 20 μ g/ml, GA: 50 μ M) combination treatment relative to the control group ($P < 0.01$; Fig. 6A). These same trends were also observed when examining TGF- β 1, Smad2, Smad3, and p-Smad2/3 protein levels ($P < 0.01$; Fig. 6B and C), with the lowest levels seen in the AST/GA group, ($P < 0.05$ or $P < 0.01$; Fig. 6B and C). Furthermore, a decrease in either mRNA or proteins of the above TGF- β 1 signaling components was noted after treatment with the TGF- β 1 inhibitor SB-431542. These results suggest that AST/GA combination treatment blocks TGF- β 1 signaling.

4. Discussion

GA was developed in China and Japan as an anti-inflammatory, antiviral, and antiallergic drug for liver disease (Li et al., 2014). A 26 weeks, randomized, phase II trial showed that GA infusion could reduce alanine transaminase levels in chronic hepatitis C (Orient et al., 2006). Our previous research showed that, AST had an effect similar to the Huangqi decoction on hepatic fibrosis induced by BDL (Mu et al., 2015). Further more, our study had indicated that the combination of AST and GA had an obvious synergistic effect in reducing liver collagen deposition and decreasing serum alanine aminotransferase (ALT) activity in DMN-induced liver fibrosis (Tong et al., 2011).

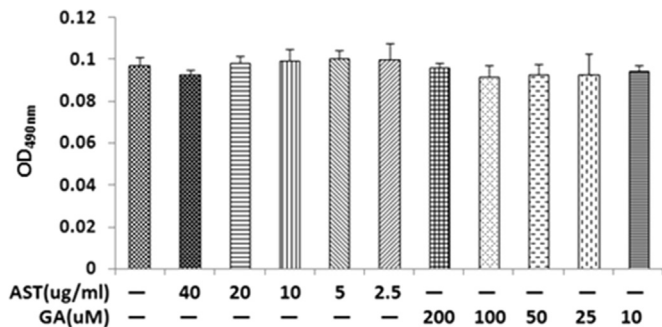


Fig. 3. AST and GA cytotoxicity in AML-12 cells.

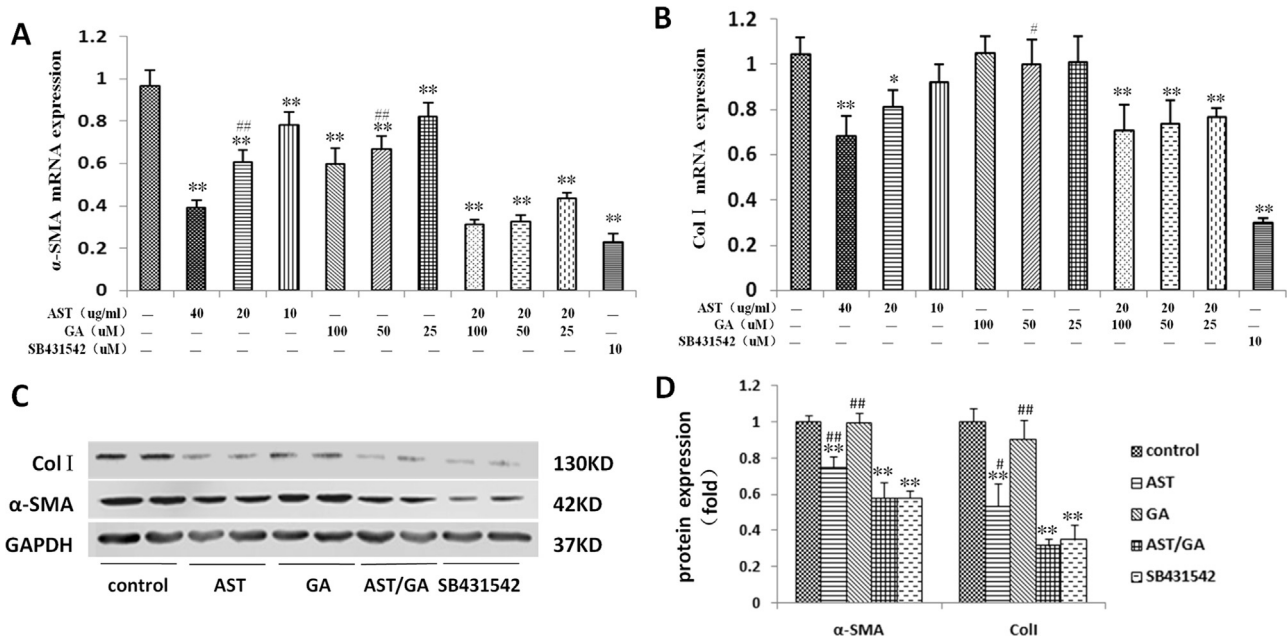


Fig. 5. Effects of AST and GA on α -SMA and Col expression in JS-1 cells. A,B: mRNA expression levels for α -SMA, Col in JS-1 cells; C,D: protein expression levels for α -SMA, Col in JS-1 cells using semiquantitative analysis. * $P < 0.05$ and ** $P < 0.01$ (vs Control); # $P < 0.05$ and ## $P < 0.01$ (vs AST/GA combination).

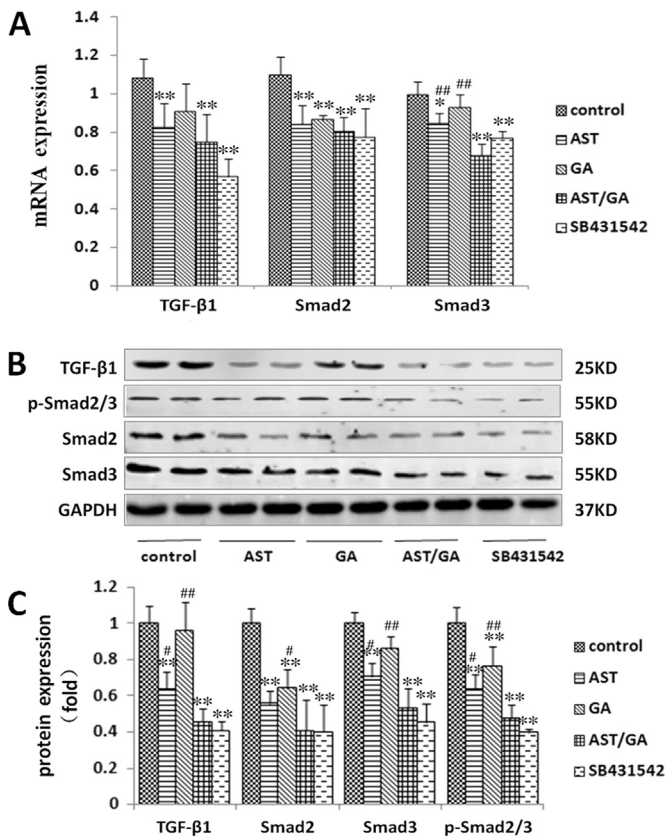


Fig. 6. Effects of AST and GA on the TGF- β 1/Smads signaling pathway in JS-1 cells. A: TGF- β 1, Smad2 and Smad3 mRNA expression. B,C: TGF- β 1, Smad2, Smad3 and p-Smad2/3 protein expression. * $P < 0.05$ and ** $P < 0.01$ (vs Control); # $P < 0.05$ and ## $P < 0.01$ (vs AST/GA combination).

In this study, BDL and DMN rat liver fibrosis models were employed to confirm that combining AST and GA can significantly improve the pathological changes associated with hepatic fibrosis, to include, significantly reducing liver Hyp levels, collagen fiber hyperplasia and serum ALT activity. In fact, the AST/GA

combination had all of these effects and was even more effective than AST or GA administered independently, making these findings consistent with our previous results (Tong et al., 2011).

Liver fibrosis results from chronic damage to the liver in conjunction with an over deposition of ECM, including fibrillar collagens, which is a characteristic of most types of chronic liver diseases. HSCs are the main ECM-producing cells in an injured liver. Following chronic injury, HSCs transdifferentiate into myofibroblast-like cells with fibrogenic properties and migrating from the space of Disse to damaged sites to aid in repair by secreting large amounts of ECM (Bataller and Brenner, 2005). Therefore, HSCs usually act as a target of therapeutic liver fibrosis, since they are potentially reversible. JS-1 cell, An immortalized mice hepatic stellate cell line, was established by Professor Scott L Friedman's group, which has the characteristics of activation itself and expression of α -SMA and Col (Guo et al., 2009; Ouyang et al., 2016). In this study, the AST/GA combination significantly inhibited JS-1 cell viability and decreased α -SMA and Col expression, with paralleled results seen both *in vitro* and *in vivo*. These results supported the synergistic actions of AST/GA and suggest that the combination of AST/GA can reverse liver fibrosis by suppressing HSC activation and ECM production.

The TGF- β 1/Smads signaling pathway plays a key role in HSC activation and the development of fibrosis (Friedman, 2008). In HSCs, TGF β 1 binds its ligand and subsequently activates R-Smads (mainly Smad2/3), a specific substrate of TGF- β 1, via phosphorylation. The phosphorylated R-Smads then associate with Smad4 to forms a complex that translocates to the nucleus to regulate the transcription of multiple target genes. Therefore, targeting of the TGF- β 1/Smads signaling pathway has become of interest as an important strategy for liver fibrosis intervention.

Our *in vitro* results indicate that the AST/GA combination remarkably reduced TGF- β 1, Smad2 and Smad3 mRNAs and protein expression, also p-Smad2/3 protein levels, with these results also noted following SB431542 treatment, a specific TGF- β 1 inhibitor. These results suggest that the AST/GA combination could reverse DMN- or BDL-induced hepatic fibrosis by suppressing HSC activation and inhibiting the TGF- β 1 signaling pathway.

Disclosures

Authors declare that they have no conflict of interest.

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